

BACK PAPER

Elementary Number Theory

Instructor: Ramdin Mawia

Marks: 45

Time: December 26, 2025; 10:00–13:00.

Attempt FIVE problems. The maximum you can score is 45.

1. Prove that the n th prime p_n is strictly less than 2^{2^n} for all $n \geq 1$. 10
2. Compute the class number $h(-147)$. 10
3. Prove that there are infinitely many primes of the form $6k + 1$. [Hint. Assume $p_1, \dots, p_n$ are all the primes of the form $6k + 1$. Look at $4(p_1 \cdots p_n)^2 + 3$ and apply the QRL.] 10
4. Solve the congruence $X^4 \equiv 9 \pmod{401}$ over $\mathbb{Z}$. 10
5. Let $(u_n)_{n \geq 0}$ be a complex sequence satisfying a linear recurrence of order 2. Suppose $u_0 = 1, u_1 = 2, u_2 = 7, u_3 = 23$. Find the n th term u_n of the sequence in closed form. 10
6. Prove that the sum of the reciprocals of the primes diverges. 10
7. Evaluate the sum 10

$$\sum_{n \leq x} \frac{1}{n}$$

with an error term which tends to 0 as $x \rightarrow \infty$.